

Bookstore Full-Stack Platform

Local Development Environment — Setup Log

Target stack: .NET 10 / C# 14 • SQL Server 2025 • Angular • OS: Windows

Setup Status: Complete

Tool	Version / Detail	Status
1. .NET SDK	10.0.400	Installed — confirmed via <code>dotnet --version</code>
2. Visual Studio Community 2026	.NET desktop dev + ASP.NET/web dev workloads	Installed
3. SQL Server 2025 Standard Developer Edition	Standard Developer (not Enterprise)	Installed
4. SQL Server Management Studio	SSMS 22, Core Components only	Installed
5. Git for Windows	2.46.1.windows.1	Already present — confirmed via <code>git --version</code>

Deliberately deferred (not installed now, per learning-strategy.md's build-first / no-front-loading rule):

- Docker Desktop — needed only for Stage J (containerize/deploy, stretch/optional)
- Node.js / Angular CLI — needed only for Stage E (Angular meets the real API); likely already present from prior work

Download Sources

Tool	Download page
.NET 10 SDK	https://dotnet.microsoft.com/en-us/download/dotnet/10.0
Visual Studio 2026 Community	https://visualstudio.microsoft.com/downloads/
SQL Server 2025 Downloads	https://www.microsoft.com/en-us/sql-server/sql-server-downloads
SQL Server Management Studio (SSMS)	https://learn.microsoft.com/en-us/ssms/install/install
Git for Windows	https://git-scm.com/download/win

Setup Steps — Detail

1. .NET 10 SDK

Downloaded the SDK (not Runtime-only) — the SDK bundles the runtime, CLI, and build tools. x64 installer, confirmed correct for standard Intel/AMD Windows machines (Arm64 only needed for ARM-based devices like Surface Pro X or Snapdragon Copilot+ PCs). Verified post-install with **dotnet --version** → reported **10.0.400**.

2. Visual Studio Community 2026

Free for individual/student use (a license condition, not a trial). Workloads selected: **.NET desktop development** and **ASP.NET and web development** — the desktop workload is required for the Console App project template

(Stage A's first build) to appear at all; the web workload covers Web API, EF Core, and NuGet from Stage C onward. All other offered workloads (Azure development, AI Development, Python, C++, Node.js, mobile, game dev, data science) were explicitly skipped — none are used anywhere in the Bookstore roadmap. VS 2026 was required over VS 2022 because 2022 cannot fully use .NET 10 / C# 14 tooling in the editor.

3. SQL Server 2025 Standard Developer Edition

Selected Developer edition over Express (Express caps database size at 10GB and strips features Stage G needs — execution plans, indexing). Within Developer edition, selected **Standard Developer** over Enterprise Developer — SQL Server 2025 split Developer into two variants for the first time, matching either the Standard or Enterprise feature set. Standard Developer was chosen because it matches what most real production environments actually run, avoiding a false sense of what's usable that Enterprise-only features would create. Nothing in the roadmap's SQL work requires an Enterprise-only feature. Basic install type, default instance.

4. SQL Server Management Studio (SSMS) 22

Installed via the shortcut offered at the end of the SQL Server installer (same tool as a direct download, fewer steps). SSMS 22 is the current, only-supported release and built for full compatibility with SQL Server 2025. Installed Core Components only — all five optional add-ons (AI Assistance, Business Intelligence, Hybrid and Migration, Code tools, Database DevOps Preview) were skipped as unrelated to the roadmap's actual SQL work. Note: Azure Data Studio was retired in February 2026 — SSMS is the correct primary tool on Windows, not ADS.

5. Git for Windows

Found already present on the machine — **git version 2.46.1.windows.1**, confirmed via **git --version**. No fresh install needed. Version is a couple of years old but fully sufficient; the operations Section 11 (Stage A close-out) requires — init/clone/commit/push/pull, branching, merge conflict resolution — have been stable across Git versions for years. Will be used to push the new BookstoreApp project to a fresh GitHub repository, kept separate from the Learning_with_AI reference repo.

Doubts Raised During Setup

Doubt 1 — Which .NET 10 SDK installer variant?

Question: “I need to download the x64 Windows installer right?”

Answer: Yes — x64 is correct for standard Intel/AMD machines. Only ARM-based Windows devices (Surface Pro X, some Snapdragon Copilot+ PCs) need Arm64. x86 (32-bit) is legacy and not relevant. Confirmed via Settings → System → About → System type, and afterward via `dotnet --version`.

Doubt 2 — VS 2026 Community download not starting in Chrome

Question: Clicking “Free Download” for Visual Studio Community in Chrome did not start the download, even after retrying via the “Click Here to retry” link.

Answer: Opened the same download link in Microsoft Edge instead — download started and completed successfully on the first try. Browser-specific issue on this machine, not a problem with the installer link itself.

Doubt 3 — Should other VS workloads be selected? (Azure and AI Development, .NET Desktop development)

Question: “Shouldn't I select the other Workloads here? Azure and AI Development, .Net Desktop development and any other?”

Answer: Correction to earlier advice: .NET desktop development should be added — Microsoft's own docs confirm the Console App template (Stage A) requires it; ASP.NET and web development alone does not surface it. Azure development and AI Development were both skipped as unrelated to the roadmap. All other offered workloads (Python, C++, Node.js, mobile, game dev, data science) were skipped too. Final set: .NET desktop development + ASP.NET and web development.

Doubt 4 — Visual Studio Build Tools showing in VS Installer, setup paused

Question: “My Visual Studio Installer has Visual Studio Build Tools, should I delete this? Its setup is currently paused.”

Answer: Not needed for this project — safe to cancel. VS Build Tools is a separate, IDE-less product for CI/build-server use; Community 2026 already includes everything it offers via the two workloads already selected. Since the install was paused (not completed), there was nothing to uninstall — just cancel the paused entry. Caveat noted: if some unrelated program on the machine needs Build Tools specifically (e.g. Node.js native module compilation via node-gyp), canceling would affect that — not flagged as a concern here since nothing like that was mentioned.

Doubt 5 — SQL Server 2025 Developer: Standard or Enterprise Developer edition?

Question: Screenshot of Microsoft's SQL Server downloads page showing two links under “SQL Server 2025 Developer”: “Download Standard Developer edition” and “Download Enterprise Developer edition.” Which one?

Answer: Standard Developer edition. SQL Server 2025 split Developer into two variants for the first time; pre-2025, Developer always meant free Enterprise, which could create a mismatch between local dev/test and real Standard-edition production. Standard Developer matches what most real production environments actually run, and nothing in the roadmap's SQL work needs an Enterprise-only feature. This also matches the priority already set for the learning process: staying compatible with what real production codebases use.

Doubt 6 — “Install SSMS” button in SQL Server installer's final popup

Question: SQL Server 2025 Standard Developer install succeeded; the last popup had an “Install SSMS” button — click it?

Answer: Yes. Same SSMS already planned as its own setup step — the SQL Server installer just offers a shortcut to it instead of a separate download page. No difference in outcome, just fewer steps.

Doubt 7 — SSMS 22 installer — correct version?

Question: The “Install SSMS” button navigated to a page offering the SSMS 22 installer — install this version?

Answer: Yes. SSMS 22 is the current, only-supported release (Microsoft dropped support for SSMS 21 and earlier once 22 shipped) and is built for full compatibility with SQL Server 2025. Caveat noted: SSMS 22's official system requirements page lists Windows 11 + Server 2019/2022/2025 only, though Microsoft's own SSMS FAQ elsewhere says “Windows 10 or later” — flagged in case a compatibility warning appeared during install on Windows 10.

Doubt 8 — SSMS 22 install: which optional components/workloads to select?

Question: Screenshot of the SSMS 22 installer's Workloads screen showing five optional add-ons (AI Assistance, Business Intelligence, Hybrid and Migration, Code tools, Database DevOps Preview), all unchecked, with SSMS Core Components already included. Which to select?

Answer: None — Core Components alone covers everything Stage B/D/G need (query editor, execution plans, index/table designer). AI Assistance, Business Intelligence, Hybrid and Migration, Code tools, and Database DevOps (Preview, and still unstable) were all skipped as unrelated to the roadmap's actual SQL work, using the same minimal-footprint principle already applied to the VS 2026 workload selection.

Generated from the Backend Learning project's ongoing setup conversation. Next step: begin Stage A (Section 1 Hands-On) using BUILD - Stage A.